Week 4-3

Discrete Random Variable: Variance

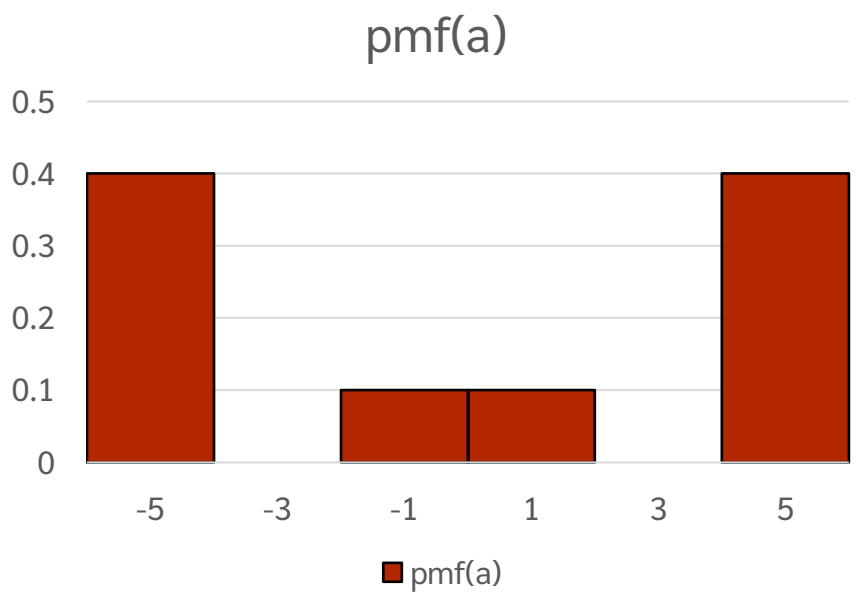
August 29, 2024

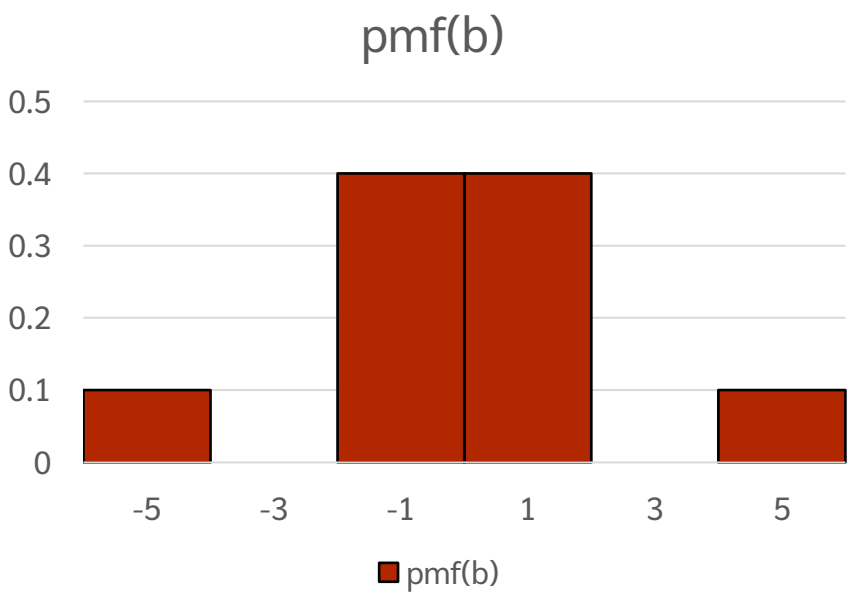
2110205 STATS COMP ENG

CHULA
COMPUTER
ΣNGINEERING

Spread

- **Expected Value** tell us the **Location Tendency.**
- **Variance** tell us **Spread** or how much data spread from the expected value.





Attendance Check

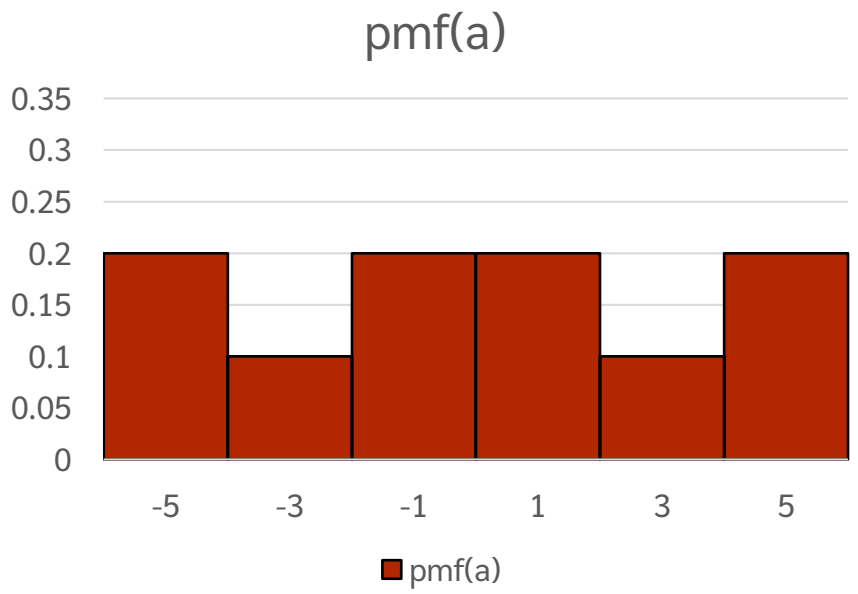
Question: Which data has higher variance?

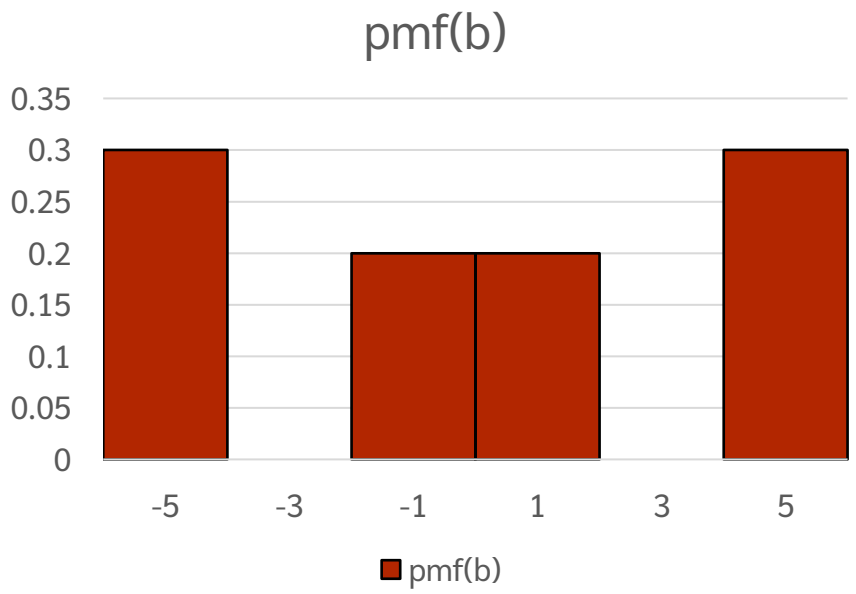
A

A	-5	-3	-1	1	3	5
p(a)	2/10	1/10	2/10	2/10	1/10	2/10

B

B	-5	-3	-1	1	3	5
p(a)	3/10	0	2/10	2/10	0	3/10





Variance and Standard Distribution

Definition: Variance of random variable X with $E(X) = \mu$ is:

$$Var(X) = \sigma^2 = E\big((X - \mu)^2\big).$$

Standard deviation σ of X is:

$$\sigma = \sqrt{Var(X)}.$$

Simpler term

- $E\left((X-\mu)^2\right)$ looks scary.
- But we can rewrite: $E\left((X-\mu)^2\right) = \boxed{\sum_{i=1}^{n} p(x_i)(x_i-\mu)^2}$

Example 1

A

A	-5	-3	-1	1	3	5
p(a)	2/10	1/10	2/10	2/10	1/10	2/10

B

B	-5	-3	-1	1	3	5
p(b)	3/10	0	2/10	2/10	0	3/10

Note: $E(A)$ and $E(B) = 0$

$$Var(A) = \frac{2}{10}(-5-0)^2 + \frac{1}{10} + (-3-0)^2 + \frac{2}{10}(-1-0)^2 + \frac{2}{10}(1-0)^2 + \frac{1}{10}(3-0)^2 + \frac{2}{10}(5-0)^2$$

$$Var(B) = \frac{3}{10}(-5-0)^2 + \frac{0}{10} + (-3-0)^2 + \frac{2}{10}(-1-0)^2 + \frac{2}{10}(1-0)^2 + \frac{0}{10}(3-0)^2 + \frac{3}{10}(5-0)^2$$

VAR(A) = 12.6, VAR(B) = 15.8 ➔ **VAR(A) < VAR(B)**

Example 2

Example 2. Compute mean, variance, and standard deviation of the following random variable.

X	1	3	5
p(a)	**2/4**	**1/4**	**1/4**
$(x-\mu)^2$	2.25	0.25	6.25

Answer: First, calculate the mean: $\mu = 2.5$.

Then, we can calculate $(x-\mu)^2$ for all value.

We get variance: $Var(x) = \frac{2}{4} \times 2.25 + \frac{1}{4} \times 0.25 + \frac{1}{4} \times 6.25 = \mathbf{2.75}$.

And standard deviation of $\sqrt{2.75}$.

On independence

Yes, we still need to know about independence.

Example: Binomial is the sum of **<u>independent</u>** Bernoulli trials.

Definition (for now): Discrete random variable X and Y are independent f :

$$P(X = a, Y = b) = P(X = a)P(Y = b),$$

for any values of a, b.

Properties of Variance

CAUTION: $E(X+Y)$ is always $E(X)+E(Y)$ but the same is not true for variance.

1. If X and Y are independent, then $Var(X+Y) = Var(X) + Var(Y)$.
2. For constants a and b, $Var(aX+b) = a^2 Var(x)$.
3. $Var(X) = E(X^2) - E(X)^2$.

Quiz 1

Problem 1: Find $Var(X + Y)$.

Variable (independent)	Mean	Variance
X	5	4
Y	3	2

Problem 2: Find $Var(X + X)$.

Quiz 1

Problem 3: Find $E(X^2)$

Variable (independent)	Mean	Variance
X	5	4
Y	3	2

Problem 4: Find $Var(2X + 3Y)$

10 MINUTES BREAK

Mean and Variance of Distributions

Bernoulli random variable

Mean: $E(X) = p$

Variance: $Var(X) = p(1-p)$

Proof:

X	0	1
pmf $p(a)$	$1-p$	p
$(X-\mu^2)$	$(0-\mu)^2 = (0-p)^2$	$(1-\mu)^2 = (1-p)^2$

$$Var(X) = (1-p)p^2 + p(1-p)^2 = (1-p)(p)(p+1-p) = (1-p)(p)$$

Binomial random variable

Mean: $E(X) = np$

Variance: $Var(X) = \boldsymbol{n} \times p(1-p)$

➔ Binomial is just n independent Bernoulli.

RECALL: If X and Y are independent, then

$$VAR(X+Y) = Var(X) + Var(Y)$$

Uniform distribution

Mean: $E(X) = \frac{n+1}{2}$

$$\mu = \sum_{i=1}^{n} \frac{1}{n} \times i = \frac{n+1}{2}$$

Variance: $Var(X) = \frac{n^2-1}{12}$

$$\sigma^2 = \sum_{i=1}^{n} i^2 - \mu^2$$
$$= \frac{1}{n} \times \frac{n(n+1)(2n+1)}{6} - \left[\frac{n+1}{2}\right]^2$$
$$= \frac{2(n+1)(2n+1)}{12} - \frac{3(n+1)^2}{12}$$
$$= \frac{(n+1)[(4n+2)-(3n+3)]}{12}$$
$$= \frac{(n+1)(n-1)}{12} = \frac{(n^2-1)}{12}$$

Geometric distribution

Mean: $E(X) = \frac{1-p}{p}$

Variance: $Var(X) = \frac{1-p}{p^2}$

Summary of distribution

Distribution	Range X	pmf $p(a)$	Mean $E(X)$	Variance $Var(X)$
Bernoulli(p)	0, 1	$p^a(1-p)^{1-a}$	p	$p(1-p)$
Binomial(n,p)	0, 1, ..., n	$\binom{n}{a}(1-p)^{n-a}p^a$	np	$np(1-p)$
Uniform(n)	1, 2, ..., n	$\frac{1}{n}$	$\frac{n+1}{2}$	$\frac{n^2-1}{12}$
Geometric(p)	0, 1, 2, ...	$(1-p)^a p$	$\frac{1-p}{p}$	$\frac{1-p}{p^2}$

Tip on using specific distribution

Last time: The probability that you pass this class is 0.9. On average, **<u>how many time you would fail</u>** before passing this course?

New question: The probability that you pass this class is 0.9. On average, **<u>how many time you need to take the course</u>** before passing?

Answer: $X \sim Geometric(0.9)$ **BUT** we want $E(X+1)$.

(which is $E(X) + 1 =$1.111....)

A topic on Memoryless-ness

A geometric distribution is **memoryless**.

➔Previous result **DOES NOT** affect new result.

$$\Pr(X > m + n | X \geq m) = Pr(X > n)$$

Example: The chance of you wining lootbox (with no guarantee) does not change with the number of tries.

In another word: $E(X)$ does not change with the number of tries.

Quiz 2

Problem 1: We toss a 100-face dice with face 2, 4, 6, ..., 200. What is the mean and variance of the value of the dice.

Problem 2: 1,000 people lined up to play at a slot machine. The machine has a success rate of 0.2%. Each person can only play once. 300 people has played unsuccessfully. On average, how many more person will fail until we get a winner?

A Fun Problem

Example: 18% of the population has a positive result on the COVID-19 test with 90% accuracy. 1,000 people with negative test attended an event. On average, how many people has COVID-19?

Answer: Let A be an event that a person has COVID and B be a positive test.

We wanted to know $P(A|B') = \frac{P(B'|A)P(A)}{P(B')}$.

P(B|A) = 0.9 P(B|A') = 0.1 P(B) = 0.18

➔ P(A) = 0.1 ➔ $P(A|B') = \frac{(1-P(B|A))P(A)}{(1-P(B))} = \frac{(1-0.9)(0.1)}{(1-0.18)} \approx 0.0122$

Number of people is $E\big(Binomial(1000,0.0122)\big) \approx 12.2$ persons.

Another Fun Problem

Example: A casino has a game. If you throw 2 on a dice, then you can play a roulette. If you win a roulette, then you get a prize. 500 persons play the game with 20 people winning. What are the chance of winning the roulette?

Answer: We have $E\big(Binomial(500, p)\big) = 500p = 20$. We get $p = 0.04$.

The chance of winning both game is $p_1 \times \mathrm{p}_2 = 0.04$. We know $p_1 = \frac{1}{6}$. Thus, $p_2 = 0.04 \div \frac{1}{6} = 0.24$.

More Fun Problem

Problem: Given a (independent) stock of NVIDIA and CPF.

NVIDIA	-2	-1	0	1
pmf	0.1	0.2	0.1	0.6

CPF	-1	0	+1
pmf	0.3	0.3	0.4

1) If you buy 4 NVIDIA stocks, what is the probability of getting negative return? What are the mean and variance?
2) What about CPF?
3) What if you buy 2 of each?

More Fun Problem (Solution)

NVIDIA	-2	-1	0	1
pmf	0.1	0.2	0.1	0.6

CPF	-1	0	+1
pmf	0.3	0.3	0.4

1. Pr(Lose Money) = 0.1+0.2. Mean = 4×0.2 Variance = $4^2 \times 1.16$
2. Pr(Lose Money) = 0.3. Mean = 4×0.1 Variance = $4^2 \times 0.69$
3. Mean = $2 \times 0.2 + 2 \times 0.1$. Variance = $2^2 \times 1.16 + 2^2 \times 0.69$. Pr(Lose Money) = 0.25

2NVIDIA / 2CPF	-4	-2	0	+2
-2	-6, Pr=0.03	-4, Pr=0.06	-2, Pr=0.03	0, Pr=DC
0	-4, Pr=0.03	-2, Pr=0.06	0, Pr=DC	2, Pr=DC
+2	-2, Pr=0.04	0, Pr=DC	2, Pr=DC	4, Pr=DC

FINISH

Homework

Assignment 4 has been updated with 2 more problems.

The required problem has not changed.

DEADLINE: MON SEP-2. NO LATE SUBMISSION.

Midterm Information

- **MIDTERM: AUG-5 2024** 13:00 – 16:00 (3 hours)
- Multiple Choices – Non-programmable calculator allowed.
- **1 A4 paper** (2-sided) allowed. This paper must also be submitted.
  - No formula will be provided in the exam.
- Exam will be in English.

- **AUG-3:** Q&A
- **AUG-5 (Morning):** No Lecture.

GOOD LUCK ON YOUR MIDTERM